

challenges in stryker:

project requirement 1:

trigger thor-windows gitlab repo pipeline and generate install-root folder and compress to zip upload to jfrog and download install-root.zip.

install-root its kind of artifactory it contains libraries and binaries

The above zip folder needs to share product security team. For their black duck scanning.

For the above requirement got below error:

**docker: Error response from daemon: hcs::CreateComputeSystem
24eab6ab74d1c903183b366ec28518d5867e14cb8da594162ed83b415576f342:
The container operating system does not match the host operating system.**

To fix that issue I have created new dockerfile to create new image:

below is the format:

Dockerfile – where I created in windows powershell

```
=====
=====
# FROM artifactory.osep.stryker.com/robotics-docker-dev/voyager/voyager-
performer:0.0.17-windows

# #FROM artifactory.osep.stryker.com/osep-docker-dev/strykercorp/osep/osep-
msvc:17-windows-ltsc2019

# #FROM artifactory.osep.stryker.com/osep-docker-dev/strykercorp/osep/osep-
msvc:17-servercore-ltsc2019

ARG
WINDOWS_BASE_IMAGE=artifactory.osep.stryker.com/mcr.microsoft.com/windows
/servercore:ltsc2022

FROM ${WINDOWS_BASE_IMAGE}

# Use CMD initially (needed for VS installer)
```

SHELL ["cmd", "/S", "/C"]

=====

Install Visual Studio Build Tools

=====

ADD https://aka.ms/vs/16/release/vs_buildtools.exe C:\\temp\\vs_buildtools.exe

RUN C:\\temp\\vs_buildtools.exe --quiet --wait --norestart --nocache \

--installPath C:\\BuildTools \

--add Microsoft.VisualStudio.Workload.MSBuildTools \

--add Microsoft.VisualStudio.Workload.VCTools -- includeRecommended \

--add Microsoft.VisualStudio.Component.VC.ATL \

--add Microsoft.VisualStudio.Component.VC.ATLMFC \

--add Microsoft.VisualStudio.Component.VC.CLI.Support \

--add Microsoft.VisualStudio.Component.VC.Modules.x86.x64 \

--add Microsoft.VisualStudio.Component.VC.Modules.ARM64 \

--add Microsoft.VisualStudio.Component.VC.Tools.ARM64 \

--add Microsoft.VisualStudio.Component.VC.Modules.ARM \

--add Microsoft.VisualStudio.Component.VC.Tools.ARM \

|| IF "%ERRORLEVEL%"=="3010" EXIT 0

SHELL ["powershell.exe", "-ExecutionPolicy", "Bypass", "-Command"]

ARG ARTIFACTORY_USER

ARG ARTIFACTORY_KEY

ARG QTCREATOR_VERSION="4.14.3"

ARG QT_VERSION="5.12.12"

```
ARG QT_INSTALL_DIR="C:/Programs/Qt"
```

```
ARG QTCREATOR_INSTALL_DIR="C:/Programs/QtCreator"
```

```
ENV ARTIFACTORY_USER=${ARTIFACTORY_USER}
```

```
ENV ARTIFACTORY_KEY=${ARTIFACTORY_KEY}
```

```
ENV QTCREATOR_VERSION=${QTCREATOR_VERSION}
```

```
ENV QT_VERSION=${QT_VERSION}
```

```
ENV QT_INSTALL_DIR=${QT_INSTALL_DIR}
```

```
ENV QTCREATOR_INSTALL_DIR=${QTCREATOR_INSTALL_DIR}
```

```
# =====
```

```
# Fix Chocolatey source credentials
```

```
# =====
```

```
#RUN choco source add --name=artifactory --
```

```
source=https://artifactory.osep.stryker.com/artifactory/api/nuget/v2/chocolatey-org --
```

```
user=$env:ARTIFACTORY_USER --password=$env:ARTIFACTORY_KEY --
```

```
priority=1 --force; choco source list
```

```
RUN powershell -ExecutionPolicy Bypass -Command "Set-ExecutionPolicy Bypass -  
Scope Process -Force; [System.Net.ServicePointManager]::SecurityProtocol = 3072;  
iex ((New-Object  
System.Net.WebClient).DownloadString('https://community.chocolatey.org/install.ps1  
'))"
```

```
RUN choco --version
```

```
# =====
```

```
# Install VS 2014 Build Tools (MSVC 15 / v141 toolset)
```

```
# =====
```

```
#COPY vs_BuildTools.exe C:/Programs/
```

```
#RUN powershell -NoProfile -ExecutionPolicy Bypass -Command "New-Item -  
ItemType Directory -Force C:\Programs | Out-Null; Invoke-WebRequest -Uri
```

```
'https://artifactory.osep.stryker.com/artifactory/dre-generic-
dev/tools/installer/windows/x86_64/MSVC/2014/BuildTools/vs_BuildTools.exe' -
Headers @{ 'X-JFrog-Art-API' = $env:ARTIFACTORY_KEY } -OutFile
C:\Programs\vs_BuildTools.exe; if (!(Test-Path C:\Programs\vs_BuildTools.exe))
{ throw 'Download failed' }; C:\Programs\vs_BuildTools.exe --quiet --wait --norestart -
-nocache --installPath C:\BuildTools --add Microsoft.VisualStudio.Workload.VCTools
--add Microsoft.VisualStudio.Component.VC.v140.x86.x64 --includeRecommended"

#RUN choco install visualstudio2019buildtools -y
```

```
#RUN powershell -Command "Start-Process C:\Programs\vs_BuildTools.exe -
ArgumentList '/Passive','/Restart','/Full' -Wait;
```

```
# Verify MSVC 15 toolset installed
```

```
RUN if (-not (Test-Path 'C:\BuildTools\VC\Tools\MSVC\14..30133')) { throw 'MSVC14
not found' }
```

```
# =====
```

```
# Python 3.11 (already in base image)
```

```
# =====
```

```
RUN choco install -y --no-progress python3 --version 3.11.6
```

```
RUN py -3.11 --version
```

```
# =====
```

```
# Supporting tools
```

```
# =====
```

```
RUN choco install -y --no-progress python3 --version 3.11.6
```

```
#RUN choco install -y python2
```

```
RUN choco install -y 7zip
```

```
#RUN choco install -y python --version=3.8.0
```

```
RUN python -m pip install --no-cache-dir --upgrade pip
```

```
RUN pip install --upgrade setuptools
```

```
RUN pip install --upgrade gcloud
```

```
RUN pip install pycryptodome
```

RUN pip install pyrebase4

RUN pip install urllib3

RUN pip install aqtinstall

=====

Perforce — already in base image, verify only

=====

#RUN p4 -V

=====

Python packages for Qt install

=====

RUN py -3.11 -m pip install --no-cache-dir --upgrade pip setuptools wheel

RUN py -3.11 -m pip install --no-cache-dir aqtinstall

=====

Install Qt 5.12.12 for MSVC 2017 x64

— Qt 5.12 is the LTS that shipped with VS 2017 support

— matches Qt Creator 4.x era

=====

RUN New-Item -ItemType Directory -Force -Path \$env:QT_INSTALL_DIR | Out-Null;

py -3.11 -m aqt install-qt --outputdir \$env:QT_INSTALL_DIR windows desktop

\$env:QT_VERSION win64_msvc2015_64

Verify qmake and add to PATH

RUN \$qmake = (Get-ChildItem -Path \$env:QT_INSTALL_DIR -Recurse -Filter
'qmake.exe' | Select-Object -First 1).DirectoryName; if (-not \$qmake) { throw
'qmake.exe not found after Qt install' };

[Environment]::SetEnvironmentVariable('PATH', \$env:PATH + ';' + \$qmake,

[EnvironmentVariableTarget]::Machine)

RUN qmake --version

RUN New-Item -ItemType Directory -Force -Path C:/Programs/Qt/Online

=====

Install Qt Creator 4.14.3 from Artifactory

— Last Qt Creator release fully supporting Qt 5.12 + MSVC 2017

=====

RUN New-Item -ItemType Directory -Force -Path C:/temp | Out-Null; \$url =
'https://artifactory.osep.stryker.com/artifactory/robotics-maven-
dev/com/strykercorp/windows/x86_64/QtCreator-StrykerEdition/' +
\$env:QTCREATOR_VERSION + '/QtCreator-StrykerEdition-' +
\$env:QTCREATOR_VERSION + '.exe'; Write-Host ('Downloading QtCreator from ' +
\$url); curl.exe --fail --silent --show-error -H ('X-JFrog-Art-API:' +
\$env:ARTIFACTORY_KEY) -o C:/temp/qtcreator.exe \$url; if (\$LASTEXITCODE -ne
0) { throw ('curl failed with exit code ' + \$LASTEXITCODE) }

RUN New-Item -ItemType Directory -Force -Path \$env:QTCREATOR_INSTALL_DIR
| Out-Null; Start-Process -FilePath C:/temp/qtcreator.exe -ArgumentList '/quiet',
'/norestart', ('INSTALLDIR=' + \$env:QTCREATOR_INSTALL_DIR) -Wait -
NoNewWindow; Remove-Item C:/temp/qtcreator.exe -Force

Add Qt Creator to PATH

RUN [Environment]::SetEnvironmentVariable('PATH', \$env:PATH + ';' +
\$env:QTCREATOR_INSTALL_DIR + '\bin', [EnvironmentVariableTarget]::Machine)

=====

Set MSVC 16 environment variables for build tools

=====

RUN [Environment]::SetEnvironmentVariable('VSINSTALLDIR',
'C:\BuildTools\VC\Tools\MSVC\', [EnvironmentVariableTarget]::Machine)

RUN [Environment]::SetEnvironmentVariable('VS140COMNTOOLS',
'C:\BuildTools\VC\Tools\MSVC\', [EnvironmentVariableTarget]::Machine)

Initialize MSVC14 environment

#ENTRYPOINT ["cmd", "/k", "C:\BuildTools\VC\Auxiliary\Build\vcvarsall.bat x64
&& powershell"]

SHELL ["cmd", "/S", "/C"]

To run above dockerfile below is the command:

```
docker build --build-arg ARTIFACTORY_USER=kasevaddi.muthyal@stryker.com --
build-arg
ARTIFACTORY_KEY=AKCpBwDbzqD8mJQC1oZxka9e6YgqWFZBF9HC6w9btdJb
5s4tGc8k5nhhihnKJMnmVjx9zWTie -t artifactory.osep.stryker.com/docker-dev/qbs-
build-tools:windows-python-2.7.16.qt5-15-10.vc16.qtc-4.14.3__10.0.19042.1889-
latest .
```

below is the new image which I created from above dockerfile:

```
artifactory.osep.stryker.com/docker-dev/qbs-build-tools:windows-python-2.7.16.qt5-
15-10.vc16.qtc-4.14.3__10.0.19042.1889-latest
```

and I replaced above image in the gitlab-ci.yml template as build_image.

```
build_image: "artifactory.osep.stryker.com/docker-dev/qbs-build-tools:windows-
python-2.7.16.qt5-15-10.vc16.qtc-4.14.3__10.0.20348.1006"
```

```
=====
=====
```

Below Script which was added in gitlab-ci.yml file to generate install-root.zip

```
# Create install-root ZIP inside container
```

```
- |
```

```
    docker run --rm -v "${PWD}:C:/temp" ${build_image} powershell -Command "if
(Test-Path 'C:\temp\install-root') { Compress-Archive -Path 'C:\temp\install-root\*' -
DestinationPath 'C:\temp\install-root.zip' -Force; Write-Host 'install-root.zip created' }
else { Write-Error 'install-root folder not found'; exit 1 }"
```

```
# Upload install-root ZIP to Artifactory
```

```
- |
```

```
    curl.exe -u
"${ARTIFACTORY_STRYKER_USER}:${ARTIFACTORY_STRYKER_KEY}" `
    -T "$CI_PROJECT_DIR\install-root.zip" `
    "https://artifactory.osep.stryker.com/artifactory/robotics-maven-
dev/tmp/$env:CI_PIPELINE_ID/install-root.zip"
```

```

docker run --rm -v "${PWD}/C:/temp" as "C:/temp" artifactory.osep.stryker.com/osep-docker-dev/stryker-corp/osep-build-tools/osep-cpp-build-tools:0.1.56 msvc15-server-lts:2022 powershell.exe 'nimgo -task:installPolicy Bypass -f ./installer_gen.ps1'
Create install-root ZIP folder container
}
docker run --rm -v "${PWD}/C:/temp" $(build_image) powershell -Command 'If (Test-Path "C:\temp\install-root") { Compress-Archive -Path "C:\temp\install-root" -DestinationPath "C:\temp\install-root.zip" -Force; Write-Host "install-root.zip created" } else { Write-Error "install-root folder not found"; exit 1 }'
Upload install-root ZIP to Artifactory
}
curl.exe -u "${ARTIFACTORY_STRYKER_USER}:${ARTIFACTORY_STRYKER_KEY}" -# collapsed multi-line command
-F "${CI_PROJECT_NAME}install-root.zip"
https://artifactory.osep.stryker.com/artifactory/robotics-maven-dev/tmp/Bom-CI_PIPELINE_ID/install-root.zip"

```

Below image was the result of install-root.zip

```

4370 install-root.zip created
4371 $ curl.exe -u "${ARTIFACTORY_STRYKER_USER}:${ARTIFACTORY_STRYKER_KEY}" -# collapsed multi-line command
4372 % Total % Received % Xferd Average Speed Time Time Time Current
4373          Dload Upload Total Spent Left Speed
4374 100 429M 0 779 100 429M 38 21.3M 0:00:20 0:00:20 --:--:-- 196
4375 {
4376   "repo": "robotics-maven-dev",
4377   "path": "/tmp/2542303753/install-root.zip",
4378   "created": "2026-05-21T04:28:21.102Z",
4379   "createdBy": "[MASKED]",
4380   "downloadUri": "https://artifactory.osep.stryker.com/artifactory/robotics-maven-dev/tmp/2542303753/install-root.zip",
4381   "mimeType": "application/zip",
4382   "size": "450417744",
4383   "checksums": {
4384     "sha1": "f50b27591ff02794ed0f9551b6575b646fb93f8a",
4385     "md5": "75b2e90ae9f90e6d289c5be4ee947795",
4386     "sha256": "6d596f7d47a26ccada6a6ec60ee64c1e08a643b289d7bd896f47f7e460561348"
4387   },
4388   "originalChecksums": {
4389     "sha256": "6d596f7d47a26ccada6a6ec60ee64c1e08a643b289d7bd896f47f7e460561348"
4390   },
4391   "uri": "https://artifactory.osep.stryker.com/artifactory/robotics-maven-dev/tmp/2542303753/install-root.zip"
4392 }
4393 Running after_script
4394 Running after_script...
4395 $ cd $CI_PROJECT_DIR
4396 $ docker run --rm -v ${CI_PROJECT_DIR}:${workdir} -w ${workdir}/modules/perforce -e PYTHONDONTWRITEBYTECODE=1 --entrypoint=python -t ${perforce_image}
4397 remove_client.py --p4port ${P4PORT} --p4user ${P4USER} --p4passwd ${P4PASSWD} --p4client ${client_name}
4397 ['remove_client.py', '-p4port', 'perforce.leibinger.strykercorp.com:1666', '--p4user', 'background', '--p4passwd', '[MASKED]', '--p4client', 'Thor-Windo
ws_']
4398 :\\Python311\\python.exe
4399 Cleaning up project directory and file based variables
4400 Job succeeded

```

Duration: 11 minutes 44 seconds

Finished: 2 days ago

Queued: 41 seconds

Timeout: 4h (from job) [🔗](#)

Runner: #34621578 (Yzis84eb)
FrioSepWindows2024G-shell-runner

Source: Web

Tags: [docker-for-windows](#) [stryker](#) [itsc2022](#)

Commit [b8cc4a39](#) [🔗](#)

Add install-root zip packaging and u
pload to Artifactory

Pipeline #2542303753 [🟢 Passed](#)
for upload_zip_folder_to_artifac
t-registry [🔗](#)

Related jobs

→ [kneet-0_build](#)